



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105



Student Name: Yoga Madha A

Reg. No.:192425058

Course Code: MLA0301

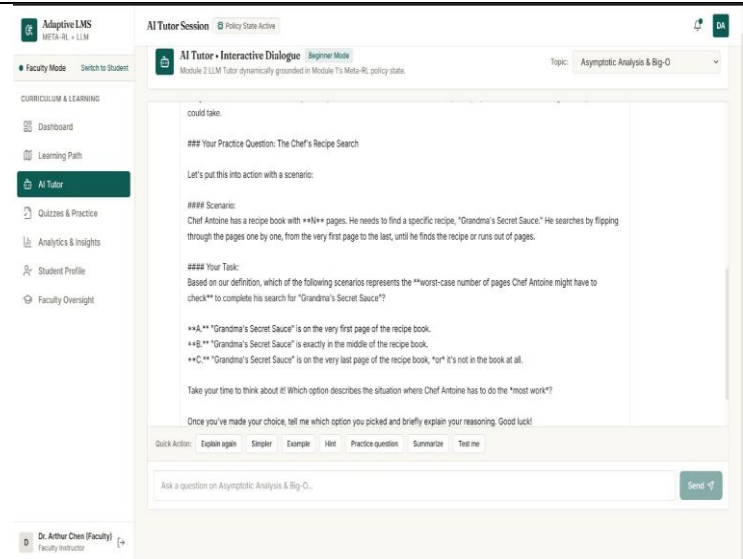
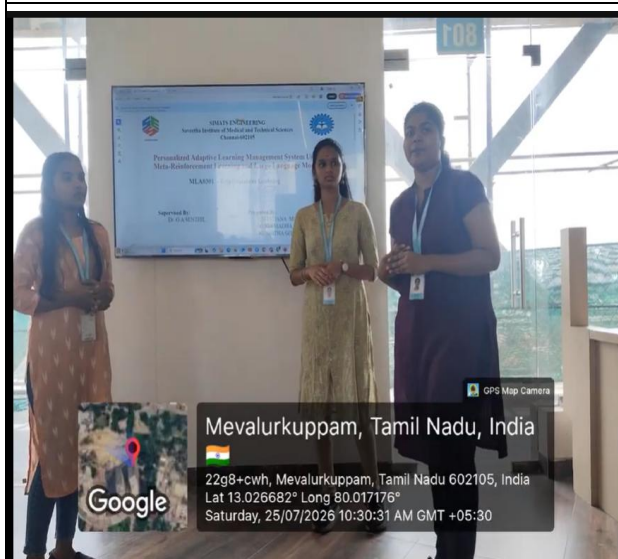
Slot: C

Course Name: Reinforcement Learning

Course Faculty: Dr.G.A.Senthil

Project Title: Personalized Adaptive Learning Management System Using Meta-Reinforcement Learning and Large Language Models

Module Photographs: Module 2 – AI Tutor Using Large Language Models



Project Description: (here you write what you did in this project (contribution) including Model Description)

Project Description:

I contributed to **Module 2 – AI Tutor Using Large Language Models**, focusing on developing the intelligent tutoring component of the Personalized Adaptive Learning Management System. I implemented an interactive AI Tutor that provides **personalized concept explanations, doubt clarification, examples, hints, summaries, and practice questions** according to the learner's current knowledge level and learning progress. The tutor uses the **student learning state and Meta-Reinforcement Learning recommendations from Module 1** to select suitable content and difficulty. I also contributed to the **adaptive quiz and instant feedback functionality**, where student responses are analyzed to identify mistakes and provide corrective explanations.

Model Description:

The AI Tutor is based on a **Large Language Model (LLM)** that generates context-aware educational responses. The model adapts explanations for **beginner, intermediate, and advanced learners** and supports interactive learning actions such as Explain Again, Simpler, Example, Hint, Practice Question, Summarize, and Test Me.

Student Signature

Guide Signature